(Please write your Exam Roll No.) *Exam Roll No. .....*

ANNUAL EXAMINATION

EIGHTH SEMESTER [B.TECH.] JULY-2023

Paper Code: ETEC-406 ***Subject: Ad Hoc and Sensor Networks***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q.No1 which is compulsory. Select one question from each unit.

Q1 Attempt All:

(a) What is an ad-hoc network? Why ad hoc networks are needed? Discuss. **(4)**

(b) What is Sensor Network Localization. **(4)**

(c) Explain goals to be achieved in transport layer protocol for ad hoc wireless networks. **(5)**

(d) List the characteristics of ideal routing protocol for ad hoc wireless network. **(4)**

(e) Define data relaying in a WSN. **(4)**

(f) What is data dissemination in a wireless sensor network? **(4)**

UNIT-I

Q2 (a) Explain about the Contention-based MAC protocols with scheduling mechanism. **(6.5)**

(b) Outline the design challenges in mobile adhoc networks and wireless sensor network. **(6)**

Q3 (a) What are the advantages of reservation based MAC protocols over contention based MAC protocols? **(6.5)**

(b) Explain the issues in designing the MAC protocols in ad hoc networks and give the classification of MAC protocols. **(6)**

UNIT-II

Q4 (a) What is the need for power management in ad hoc network? Discuss the approaches for power aware routing protocol. **(6)**

(b) Give classification of routing protocols in ad hoc networks. Explain any two. **(6.5)**

Q5 (a) Why TCP protocols used in wired network is not suitable for wireless networks? **(6.5)**

(b) What do you understand by network security requirements? Discuss the issues and challenges in secure routing of ad hoc wireless networks. **(6)**

P.T.O.